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Database Administration

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بِسْمِ اللهِ الرَّحْمٰنِ الرَّحِيمِ

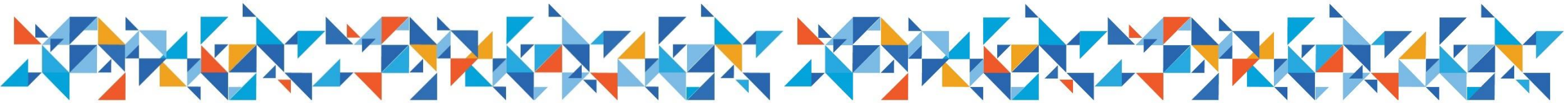
اَلْحَمْدُ لِلّٰهِ رَبِّ الْعَالَمِينَ

وَالصَّلَاةُ وَالسَّلَامُ عَلَىٰ سَيِّدِنَا مُحَمَّدٍ ﷺ

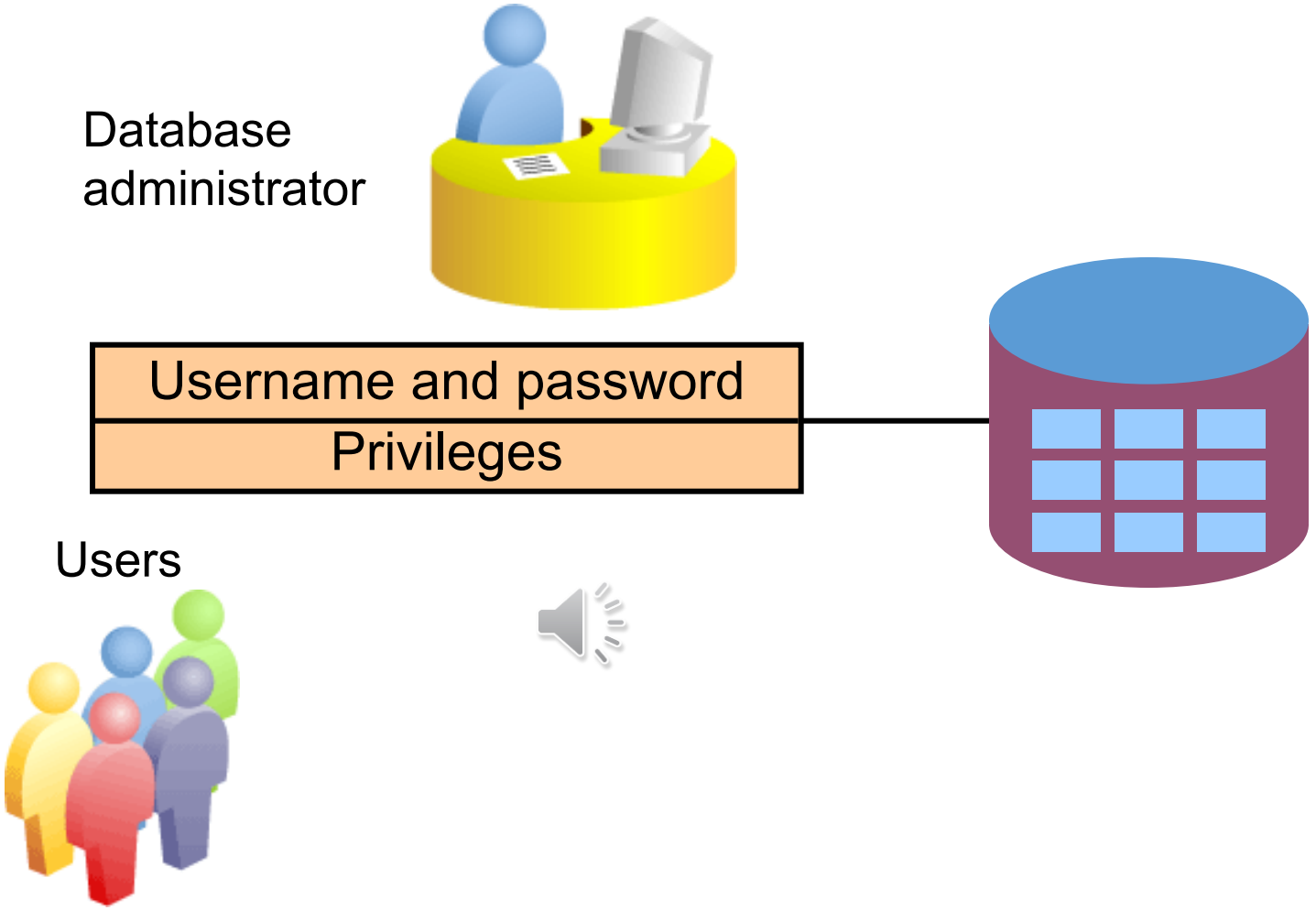
Preview

Controlling User Access

- Creating Users
- Granting System Privileges
- Users Space
- Role
- Creating and Granting Privileges to a Role
- Object Related Privileges
- Revoking Object Privileges



Controlling User Access



Privileges

- Database security:
  - System security
  - Data security
- System privileges: Performing a particular action within the database
- Object privileges: Manipulating the content of the database objects
- Schemas: Collection of objects such as tables, views, and so on.



System Privileges

- The database administrator has high-level system privileges for tasks such as:
  - Creating new users
  - Removing users
  - Removing tables



Creating Users

•The database administrator (DBA) creates users with the `CREATE USER` statement.

```
CREATE USER user
IDENTIFIED BY   password;
```



```
CREATE USER  demo
IDENTIFIED BY demo;
```

User System Privileges

- After a user is created, the DBA can grant specific system privileges to that user.

```
GRANT privilege [, privilege...]
TO user [, user| role ];
```

- An application developer, for example, may have the following system privileges:

  CREATE SESSION
  CREATE TABLE
  CREATE SEQUENCE
  CREATE VIEW
  CREATE PROCEDURE



Granting System Privileges

The DBA can grant specific system privileges to a user.

```
GRANT   create session, create table,
        create sequence, create view
TO      demo;
```



Users Space

User must have space in USER area assigned by the database administrator (DBA) in order to insert data

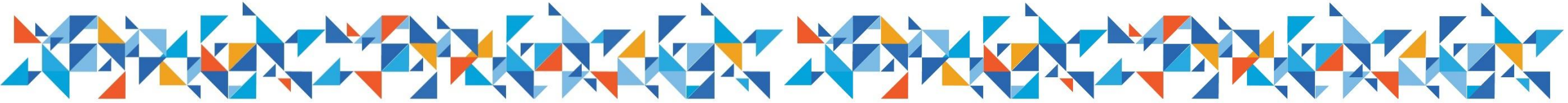


```
ALTER USER USER_NAME QUOTA 10M ON USERS;
```

Object Related Privileges

- Object privileges vary from object to object.
- An owner has all the privileges on the object.
- An owner can give specific privileges on that owner's object.

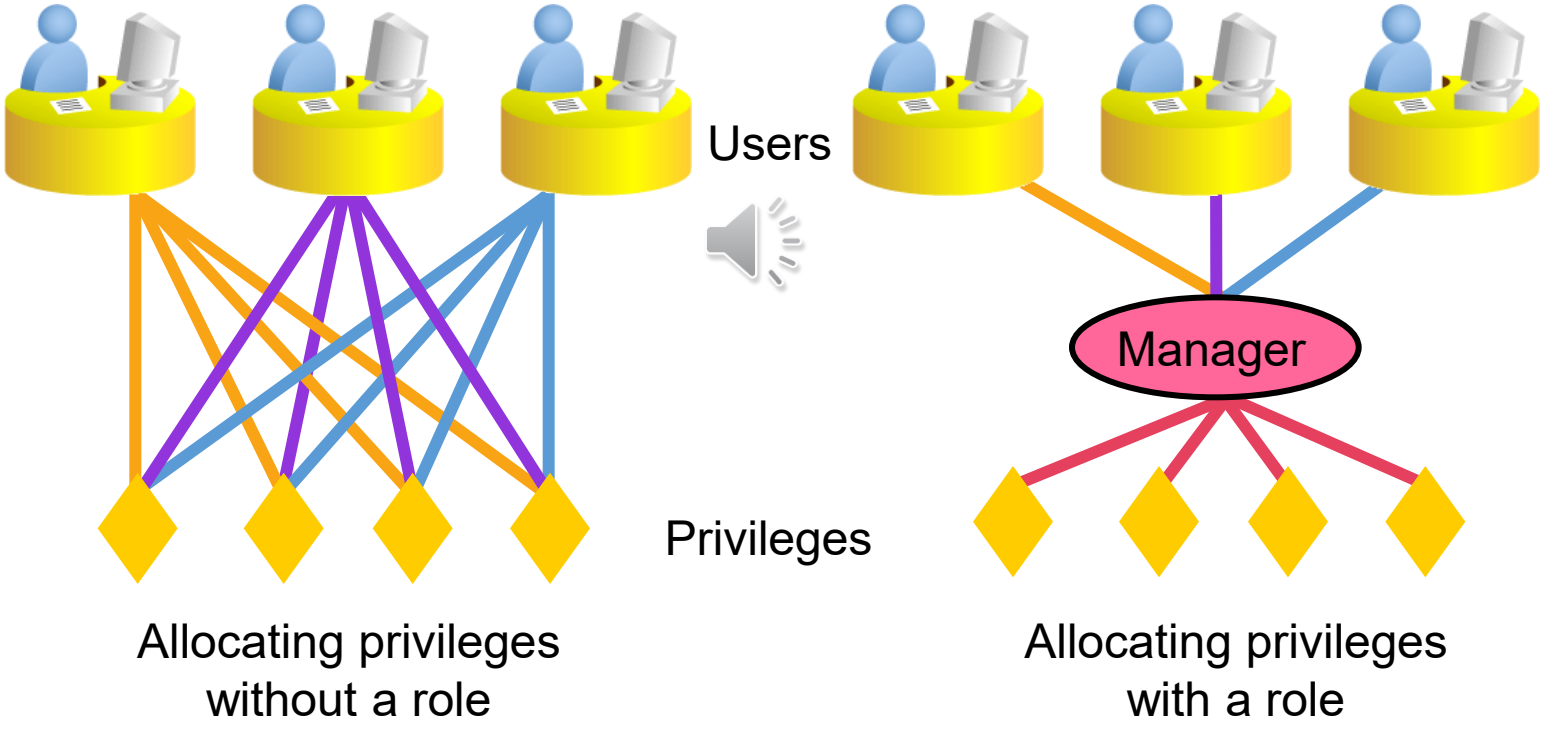
```
GRANT       object_priv
ON          object
TO          {user|role|PUBLIC}
[WITH GRANT OPTION];
```



Object Related Privileges

Object privilege	Table	View	Sequence
ALTER	✓		✓
DELETE	✓	✓	
INDEX	✓		
INSERT	✓	✓	
REFERENCES	✓		
SELECT	✓	✓	✓
UPDATE	✓	✓	

What Is a Role?



Creating and Granting Privileges to a Role

- Create a role:

```
CREATE ROLE manager;
```

- Grant privileges to a role:



```
GRANT create table, create view
TO manager;
```

- Grant a role to users:

```
GRANT manager TO BELL, Asif;
```

Granting Object Privileges

- Grant query privileges on the `EMPLOYEES` table:

```
GRANT   select
ON      employees
TO      demo;
```

- Grant privileges to update specific columns to users and roles:



```
GRANT   update (department_name, location_id)
ON      departments
TO      demo, manager;
```

Revoking Object Privileges

- You use the `REVOKE` statement to revoke privileges granted to other users.
- Privileges granted to others through the `WITH GRANT OPTION` clause are also revoked.

```
REVOKE {privilege [, privilege...]|ALL}
ON     object
FROM    {user[, user...]|role|PUBLIC};
```



Revoking Object Privileges

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- Revoke the `SELECT` and `INSERT` privileges given to the `demo` user on the `DEPARTMENTS` table.

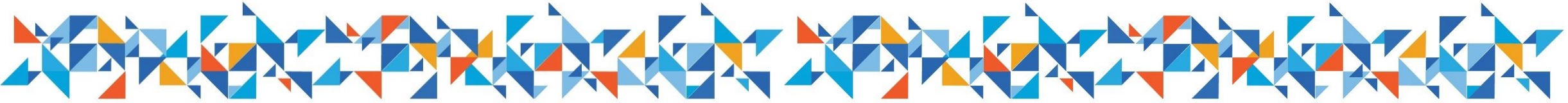
```
REVOKE   select, insert
ON       departments
FROM     demo;
```



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DBA QUIZ # 3

QUESTION: Write From Your Mind About Role Base Access Rights In Context Of ORACLE and SQL SERVER?



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Thank You